

Validation footprint heatmap — signal validity by setting (V2) p values where defined; * = point estimate only (E191 setting means)

E189 random pair	0.37-0.41	0.24			
E189 unseen column	0.21-0.25	-0.09			
E189 unseen row	-0.10--0.01	0.10*			
E189 double unseen	-0.35--0.24	-0.08			
E190 cross-study K562	0.42	0.42	0.47		
E192 cross-study RPE1	0.30 (CI×0)	0.34 (CI×0)			
E199 unseen genes (K562)	0.395 ✓	0.096 (CI×0)		0.16	
E200 context holdout (K562)	0.424 ✓	0.880 ✓	0.664 ✓	0.16	
E158 strict unseen (PRESCRIBE)					saturated
E201 four contexts (sealed)	SEALED	SEALED	SEALED	SEALED	
	disagreement / risk score	predicted magnitude	source-effect dispersion	model-baseline gap	PRESCRIBE official

VALID (CI>0)

degraded (CI×0) → ABSTAIN

undefined (saturated)

positive point only*

negative → ABSTAIN

sealed (E201)